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Hand-in 2

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Answers

- True
 - False
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 - False
- First note that the set of even numbers, $E = \{2n : n \in \mathbb{N}\} \subset \mathbb{N}$ since $\forall x \in E, x \in \mathbb{N}$ and $E \neq \mathbb{N}$ (we see that, for example, $1 \in \mathbb{N}$ but $1 \notin E$). Now we have to prove that it is not true that $|E| < |\mathbb{N}|$. We do this by proving a bijection $E \rightarrow \mathbb{N}$ exists. It is then sufficient to prove that $|E| = |\mathbb{N}|$ to prove this. Now we can write the elements of E as the list: 2, 4, 6, 8, 10, ... and this means that $|E| = |\mathbb{N}|$ and thus proving this statement false
- Suppose A and B are non-empty sets with $|A| = |B|$. This means there is a bijection $f : A \rightarrow B$. Now consider $g : \mathcal{P}(A) \rightarrow \mathcal{P}(B)$ defined as $g(X) = \{f(x) : x \in X\}$ for all $X \in \mathcal{P}(A)$. We need to prove this is a bijection. Suppose $g(X) = g(Y)$ for some $X, Y \in \mathcal{P}(A) \Rightarrow \{f(x) : x \in X\} = \{f(x) : x \in Y\}$. Now to prove $X = Y$, we prove that $X \subseteq Y$ and $Y \subseteq X$. Suppose $y \in g(X) \Rightarrow \exists x \in X, f(x) = y$. Also, $y \in g(Y) \Rightarrow \exists x' \in Y$ such that $f(x') = y = f(x) \Rightarrow x = x'$ since f is injective and so $x \in Y$. So $X \subseteq Y$. Similarly, $Y \subseteq X$, meaning $X = Y$ and so g is injective. Now suppose $Y \in \mathcal{P}(B)$. Each $y \in Y$ has some $x \in X$ such that $f(x) = y$ because f is surjective and so we can construct a set $A = \{x \in X : f(x) \in Y\}$ and then this is the preimage of Y . Then $g(A) = \{f(x) : x \in A\} = Y$ so g is surjective. Thus g is a bijection, meaning $|\mathcal{P}(A)| = |\mathcal{P}(B)|$, so this statement is true.